

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

## 4<sup>th</sup> Semester of B. Pharm. Examination University Theory Examination May 2016 PH223 Pharmaceutical Biochemistry

Date: 09.05.2016, Monday

Time: 10:00 a.m. to 1:00 p.m.

Maximum Marks: 80

### Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

### SECTION - I

- Q 1** Attempt all questions. Each question is of one mark. **20**
1. Trans amination is catalysed by \_\_\_\_\_.  
[A] Transaminase  
[B] Aminotransferase  
[C] Both of above  
[D] None of above
  2. Urea is the end product of \_\_\_\_\_ metabolism.  
[A] Protein  
[B] Amino acid  
[C] Carbohydrate  
[D] All of above
  3. All transaminases requires \_\_\_\_\_ as co enzyme.  
[A] pyridoxol phosphate  
[B] Pyridoxol carbonate  
[C] Pyridoxine phosphate  
[D] None of above
  4. Transfer of amino group from the amino acid to keto acid is known as \_\_\_\_\_.  
[A] Transamination  
[B] Transketonation  
[C] Both of above  
[D] ketogenesis
  5. \_\_\_\_\_ is involved in production of ATP.  
[A] Golgi Appratus  
[B] Mitochondria  
[C] Lysosomes  
[D] All of above
  6. Some pathways degrade organic nutrients into simple end products in order to extract chemical energy and convert it into a form useful to the cell; together these degradative, free-energy-yielding reactions are designated \_\_\_\_\_.  
[A] Catabolism  
[B] Anabolism  
[C] Both of above  
[D] None of above

7. A compound that is responsible for enhancing rate of reaction without consumed in the biochemical reaction is known as  
[A] Enzymes  
[B] Co-enzymes  
[C] Entropy  
[D] None of above
8. A programmed cell death or cell suicide is known as  
[A] Cipoptosis  
[B] Apoptosis  
[C] Artificial death  
[D] Repoptosis
9. Excess uric acid causes  
[A] Arthritis  
[B] Gout  
[C] Lesch-Nyhan syndrome  
[D] All of above
10. The combined action of an aminotransferase and glutamate dehydrogenase is referred as  
[A] Transamidation  
[B] Transdeamination  
[C] Deamination  
[D] None of above
11. Triglycerides are  
[A] esters of fatty acids  
[B] esters of glycerol with fatty acids  
[C] Esters of glycerol with aromatic fatty acids  
[D] esters
12. Saponification number is defined as  
[A] mg of KOH required to hydrolyse (saponify) one gram of fat or oil  
[B] ml of KOH required to hydrolyse (saponify) one gram of fat or oil  
[C] mg of KOH required to hydrolyse (saponify) one ml of fat or oil  
[D] ml of KOH required to hydrolyse (saponify) one ml of fat or oil
13. The biological pathway in which as such there is no utilization or consumption of ATP is \_\_\_\_\_  
[A] Glycogenolysis  
[B] Glycolysis  
[C] TCA cycle  
[D] HMP shunt
14. Which is the protein part of the enzyme?  
[A] Coenzyme  
[B] Apoenzyme  
[C] Holoenzyme  
[D] Cofactor
15. Most of the eenzymes are active in \_\_\_\_ pH range.  
[A] Alkaline pH (8-10)  
[B] Acidic pH (1-4)  
[C] Neurtal pH (6-8)  
[D] Entire pH range (1-14)
16. Enzymes show stereospecificity except \_\_\_\_ class of enzymes.  
[A] Transferase  
[B] Isomerase  
[C] Lygase  
[D] Hydrolase
17. Disulfiram used for treatment of alcoholism shows \_\_\_\_\_ type of enzyme activity.  
[A] Reversible competitive inhibition

- [B] Reversible non-competitive inhibition  
 [C] Irreversible inhibition  
 [D] Allosteric Inhibition
18. The organisms that obtain carbon from CO<sub>2</sub> are  
 [A] chemotrophs  
 [B] heterotrophs  
 [C] phototrophs  
 [D] autotrophs
19. Two sugars which differ from one another only in configuration around a single carbon atom are termed  
 [A] Epimers  
 [B] Anomers  
 [C] Optical isomers  
 [D] Stereoisomers
20. Mutarotation refers to change in  
 [A] pH  
 [B] Optical rotation  
 [C] Conductance  
 [D] Chemical properties

### SECTION – II

- Q 2** Attempt any **FOUR** of the following
- |          |                                                   |           |
|----------|---------------------------------------------------|-----------|
| <b>A</b> | Draw Glycolysis Cycle of Carbohydrate metabolism. | <b>05</b> |
| <b>B</b> | Write a note on Fatty acid synthase complex.      | <b>05</b> |
| <b>C</b> | Draw the fatty acid Oxidation cycle.              | <b>05</b> |
| <b>D</b> | Write a note on enzyme specificity.               | <b>05</b> |
| <b>E</b> | Write a note on Oxidative Phosphorylation.        | <b>05</b> |
| <b>F</b> | Write the Biosynthesis of Cholesterol             | <b>05</b> |

### SECTION – III

- Q 3** Attempt any **FOUR** of the following
- |          |                                                                                                  |           |
|----------|--------------------------------------------------------------------------------------------------|-----------|
| <b>A</b> | Define: 1] Essential fatty acid 2] Mutarotation 3] Racemic Mixture 4] Anomers 5] stereoisomerism | <b>05</b> |
| <b>B</b> | Explain Gluconeogenesis.                                                                         | <b>05</b> |
| <b>C</b> | Write a note Hexose Monophosphate Shunt.                                                         | <b>05</b> |
| <b>D</b> | Compare prokaryote and eukaryote cells                                                           | <b>05</b> |
| <b>E</b> | Discuss the inhibitors of purine synthesis                                                       | <b>05</b> |
| <b>F</b> | Discuss the salient features of transamination.                                                  | <b>05</b> |
- Q 4** Attempt any **FOUR** of the following
- |          |                                                                                                                   |                        |
|----------|-------------------------------------------------------------------------------------------------------------------|------------------------|
| <b>A</b> | Draw and Discuss the structure of DNA                                                                             | <b>05</b>              |
| <b>B</b> | [1] Explain the biosynthesis of AMP and GMP from Inosine mono phosphate.<br>[2] Explain denaturation of proteins. | <b>03</b><br><b>02</b> |
| <b>C</b> | [1] Discuss the interrelation between urea cycle and TCA cycle.<br>[2] Explain the sources of amino acid pool.    | <b>03</b><br><b>02</b> |
| <b>D</b> | Give chemical classification of amino acids with suitable examples and structures                                 | <b>05</b>              |
| <b>E</b> | Draw and explain urea cycle.                                                                                      | <b>05</b>              |
| <b>F</b> | Explain pathway for biosynthesis of catecholamines.                                                               | <b>05</b>              |